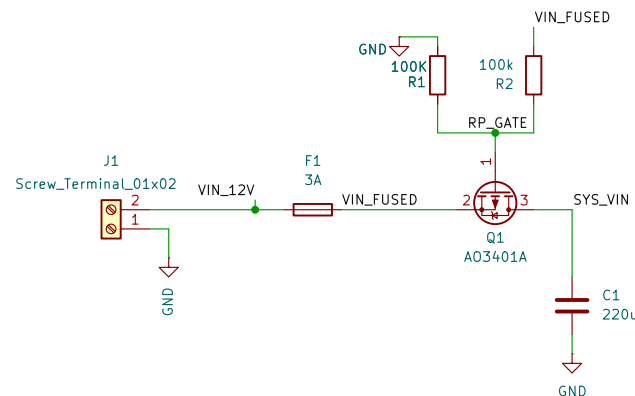
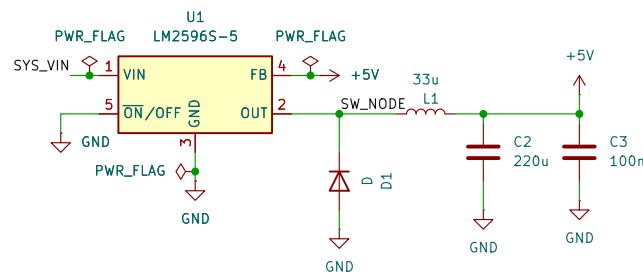


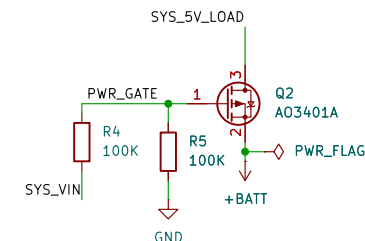
1.INPUT & protection



2.BUCK CONVERTER (12V to 5V)



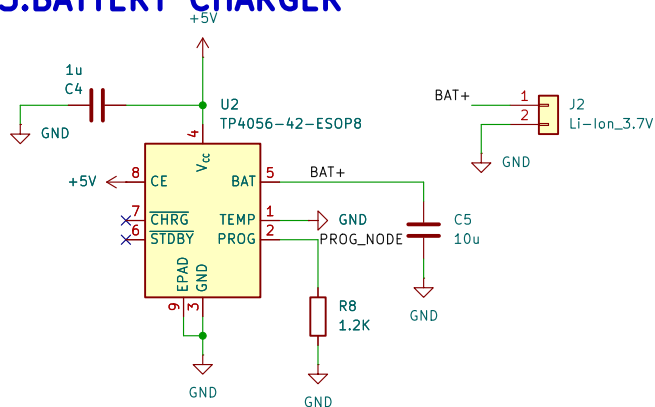
5.POWER PATH



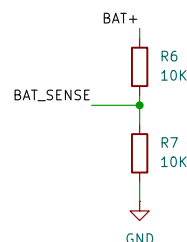
NOTES:

- Input: 12V DC
- Output: 5V
- Battery: 1S Li-Ion
- Charge current: 1A
- Max output current: 2A

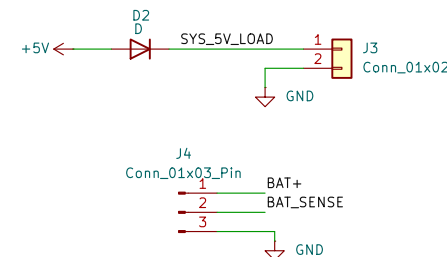
3.BATTERY CHARGER



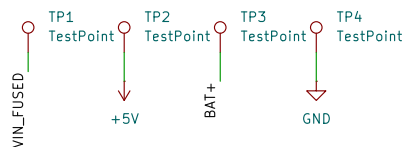
4.VOLTAGE SENSING



6.OUTPUT



TEST POINTS:



LM2596S + TP4056
12V Input - 5V Output - Li-Ion Charger
Larbi Mohamed Redha

Sheet: /
File: Smart Power Management Board..kicad_sch

Title: Smart Power Management Board

Size: A4 Date: 2026-05-08
KiCad E.D.A. 10.0.1

Rev: 1.0
Id: 1/1